

Intuition behind Matrix Multiplication

Asked 13 years, 8 months ago Modified 4 years, 5 months ago Viewed 79k times



191



What is the intuitive way of thinking about multiplication of matrices?



matrices intuition faq

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edited Aug 10, 2013 at 15:28



Jeel Shah

9,482 21 79 126

asked Apr 8, 2011 at 12:41



Happy Mittal

3,257 6 27 32

90 Do you think of $3.9289482948290348290 \times 9.2398492482903482390$ as repetitive addition? – lhf Apr 8, 2011 at 12:43

17 It depends on what you mean by 'meaning'. – Mitch Apr 8, 2011 at 12:58

29 I believe a more interesting question is *why* matrix multiplication is defined the way it is. – M.B. Apr 8, 2011 at 13:01

7 I can't help but read your question's title in Double-Rainbow-man's voice... – Adrian Petrescu Apr 8, 2011 at 16:04

4 You probably want to read this answer of Arturo Magidin's: [Matrix Multiplication: Interpreting and Understanding the Multiplication Process](#) – Utcensis Apr 8, 2011 at 16:42

14 Answers

Sorted by: Highest score (default)



148



Matrix "multiplication" is the composition of two linear functions. The composition of two linear functions is a linear function.

If a linear function is represented by A and another by B then AB is their composition. BA is the their reverse composition.

Thats one way of thinking of it. It explains why matrix multiplication is the way it is instead of piecewise multiplication.



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answered Apr 8, 2011 at 13:00



Searke

1,695 1 11 9

47 Actually, I think it's the *only* (sensible) way of thinking of it. Textbooks which only give the definition in terms of coordinates, without at least mentioning the connection with composition of linear maps, (such as my first textbook on linear algebra!) do the student a disservice. – wildildlife Apr 8, 2011 at 16:26 ✎

5 Learners, couple this knowledge with mathinsight.org/matrices_linear_transformations . It may save you a *good* amount of time. :) – n611x007 Jan 30, 2013 at 16:36 ✎

Indeed, there is a sense in which *all* associative binary operators with an identity element are represented as composition of functions - that is the underlying nature of associativity. (If there isn't an identity, you get a representation as composition, but it might not be faithful.) – Thomas Andrews Aug 10, 2013 at 15:30 ✎

4 can you give an example of what you mean? – JobHunter69 Mar 11, 2016 at 21:18

I found the Wikipedia entry on Linear Maps and specifically the Matrices section very useful to grok this: en.m.wikipedia.org/wiki/Linear_map – Angad Dec 3, 2023 at 14:54



56



Asking why matrix multiplication isn't just componentwise multiplication is an excellent question: in fact, componentwise multiplication is in some sense the most "natural" generalization of real multiplication to matrices: it satisfies all of the axioms you would expect (associativity, commutativity, existence of identity and inverses (for matrices with no 0 entries), distributivity over addition).

The usual matrix multiplication in fact "gives up" commutativity; we all know that in general $AB \neq BA$ while for real numbers $ab = ba$. What do we gain? Invariance with respect to change of basis. If P is an invertible matrix,

$$P^{-1}AP + P^{-1}BP = P^{-1}(A + B)P$$

$$(P^{-1}AP)(P^{-1}BP) = P^{-1}(AB)P$$

In other words, it doesn't matter what basis you use to represent the matrices A and B , no matter what choice you make their sum and product is the same.

It is easy to see by trying an example that the second property does not hold for multiplication defined component-wise. This is because the inverse of a change of basis P^{-1} no longer corresponds to the multiplicative inverse of P .

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edited Jul 28, 2020 at 9:49



bilibraker
75 5

answered Apr 8, 2011 at 15:41



user7530
50.1k 11 88 157

29 +1, but I can't help but point out that if componentwise multiplication is the most "natural" generalization, it is also the most *boring* generalization, in that under componentwise operations, a matrix is just a flat collection of mn real numbers instead of being a new and useful structure with interesting properties. – user856 Apr 8, 2011 at 17:09 ✎

5 I don't get it. For element-wise multiplication, we would define P^{-1} as essentially element-wise reciprocals, so PP^{-1} would be the identity, and the second property you mention would still hold. It is the essence of P^{-1} to be the inverse of P . How else would you define P^{-1} ? – Yatharth Agarwal Nov 3, 2018 at 20:27 ✎

2 I also don't get it. Under elementwise multiplication, we have commutativity. Then $P'AP = A$, a change of basis does nothing, and showing we have the two invariance properties is trivial. – Accidental Statistician Sep 8, 2021 at 23:14

1 @AccidentalStatistician Change of basis means: take a vector expressed as a linear combination in one basis, and find the linear combination in a different basis that yields the same vector. This operation is independent of how you define or "redefine" multiplication. If $\odot$ is componentwise multiplication, then matrices transform under



41



The short answer is that a *matrix* corresponds to a *linear transformation*. To *multiply* two matrices is the same thing as *composing* the corresponding linear transformations (or *linear maps*).

The following is covered in a text on linear algebra (such as Hoffman-Kunze):

This makes most sense in the context of vector spaces over a field. You can talk about vector spaces and (linear) maps between them without ever mentioning a basis. When you pick a basis, you can write the elements of your vector space as a sum of basis elements with coefficients in your base field (that is, you get explicit coordinates for your vectors in terms of for instance real numbers). If you want to compute something, you typically pick bases for your vector spaces. Then you can represent your linear map as a matrix with respect to the given bases, with entries in your base field (see e.g. the above mentioned book for details as to how). We define matrix multiplication such that matrix multiplication corresponds to composition of the linear maps.

Added (*Details on the presentation of a linear map by a matrix*). Let V and W be two vector spaces with ordered bases $e_1, \dots, e_n$ and $f_1, \dots, f_m$ respectively, and $L : V \rightarrow W$ a linear map.

First note that since the e_j generate V and L is linear, L is completely determined by the images of the e_j in W , that is, $L(e_j)$. Explicitly, note that by the definition of a basis any $v \in V$ has a unique expression of the form $a_1 e_1 + \dots + a_n e_n$, and L applied to this pans out as $a_1 L(e_1) + \dots + a_n L(e_n)$.

Now, since $L(e_j)$ is in W it has a unique expression of the form $b_1 f_1 + \dots + b_m f_m$, and it is clear that the value of e_j under L is uniquely determined by $b_1, \dots, b_m$, the coefficients of $L(e_j)$ with respect to the given ordered basis for W . In order to keep track of which $L(e_j)$ the b_i are meant to represent, we write (abusing notation for a moment) $m_{ij} = b_i$, yielding the matrix (m_{ij}) of L with respect to the given ordered bases.

This might be enough to play around with why matrix multiplication is defined the way it is. Try for instance a single vector space V with basis $e_1, \dots, e_n$, and compute the corresponding matrix of the square $L^2 = L \circ L$ of a single linear transformation $L : V \rightarrow V$, or say, compute the matrix corresponding to the identity transformation $v \mapsto v$.

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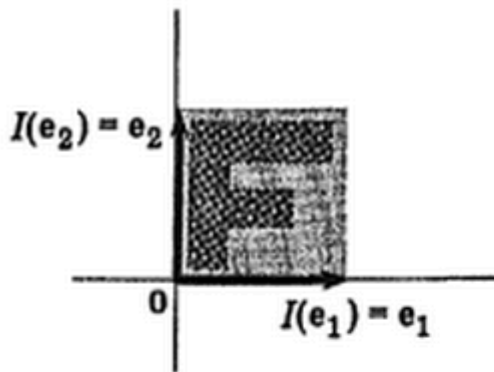
edited Jul 4, 2015 at 8:29

answered Apr 8, 2011 at 13:04

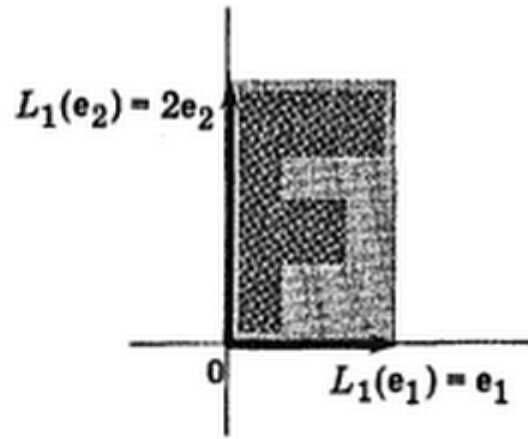


Eivind Dahl

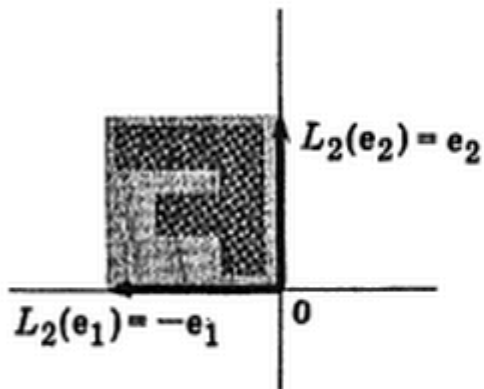
1,657 10 17



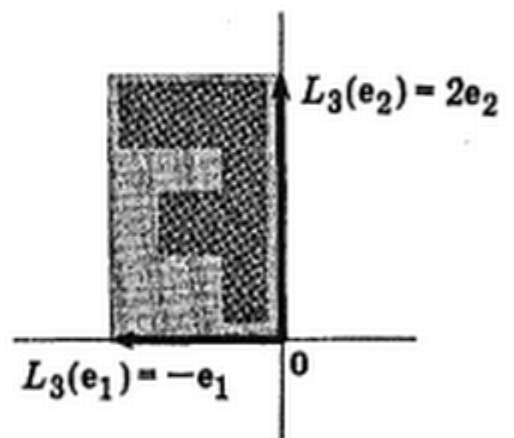
(a) Identity $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



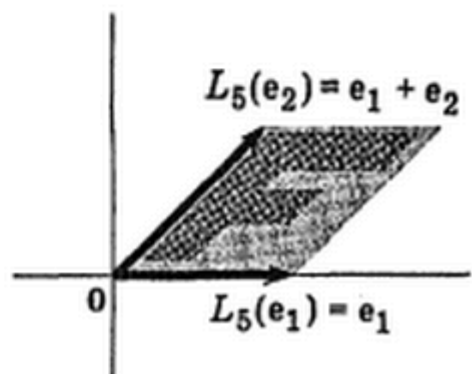
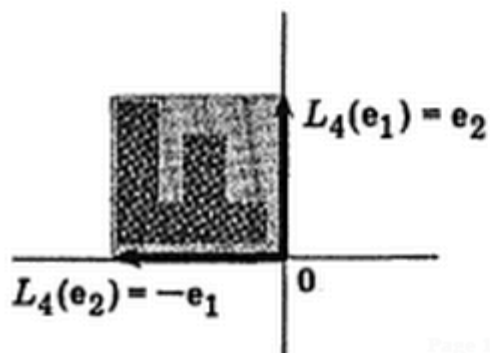
(b) Stretching $L_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

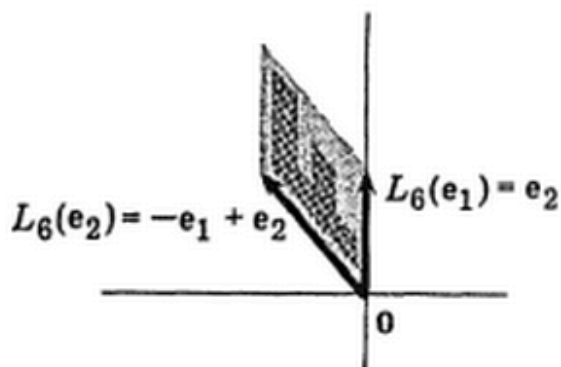


(c) Reflection $L_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

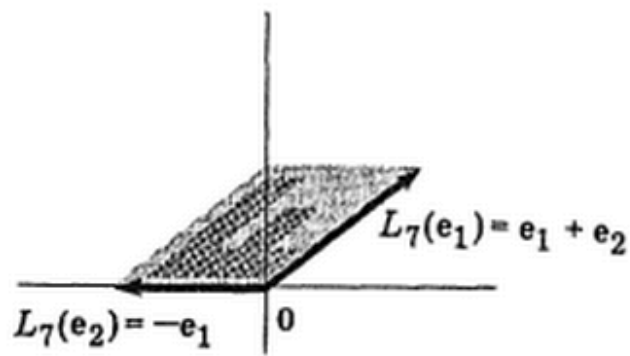


(d) Stretching and reflection $L_3 = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$

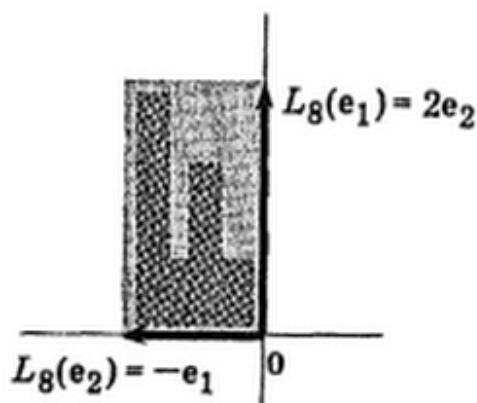




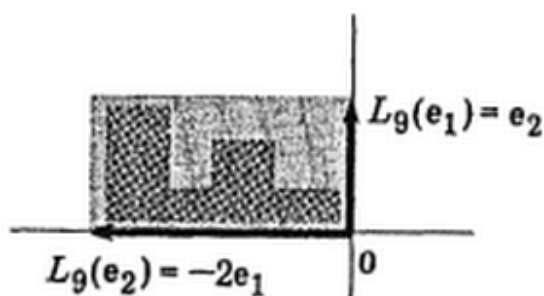
(g) Shear and rotation $L_6 = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$



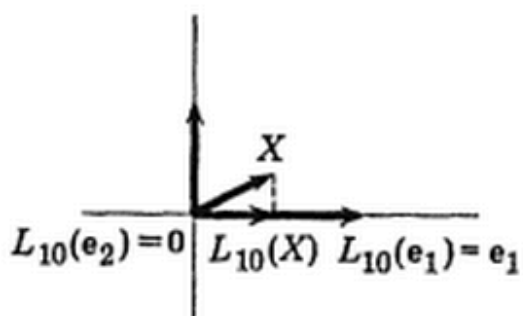
(h) Rotation and shear $L_7 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$



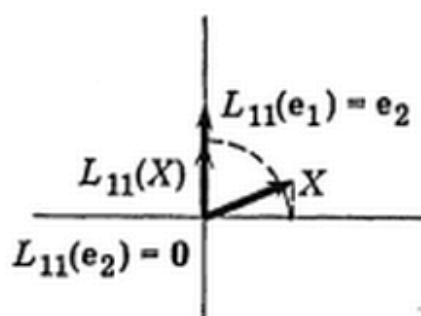
(i) Rotation and stretching
 $L_8 = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$



(j) Stretching and rotation
 $L_9 = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$

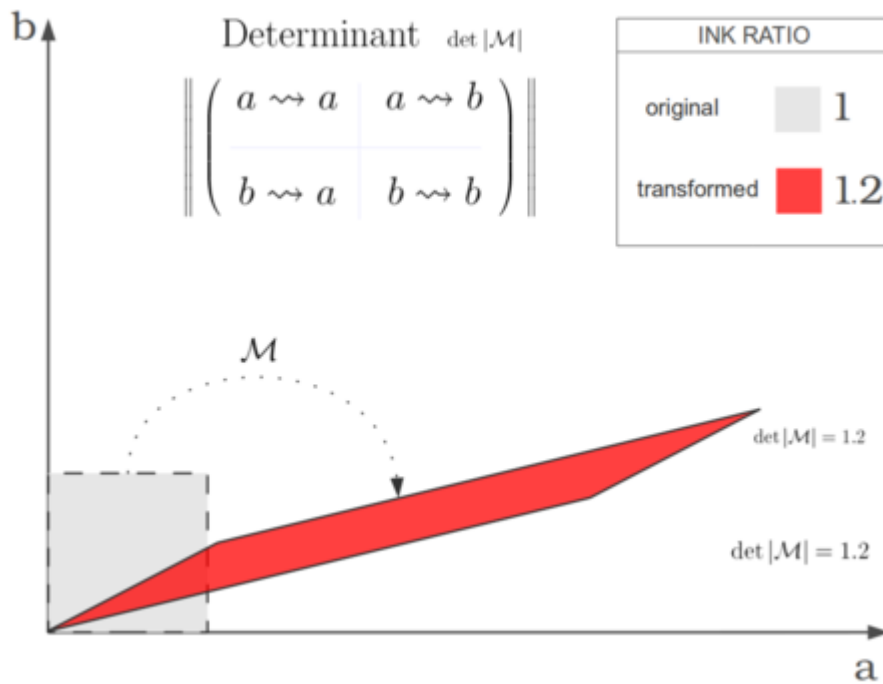


(k) Projection $L_{10} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



(l) Projection and rotation $L_{11} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

Instead of looking at a "box of numbers", look at the "total action" after applying the whole thing. It's an automorphism of linear spaces, meaning that in some vector-linear-algebra-type situation this is "turning things over and over in your hands without breaking the algebra that makes it be what it is". (Modulo some things—like maybe you want a constant determinant.)



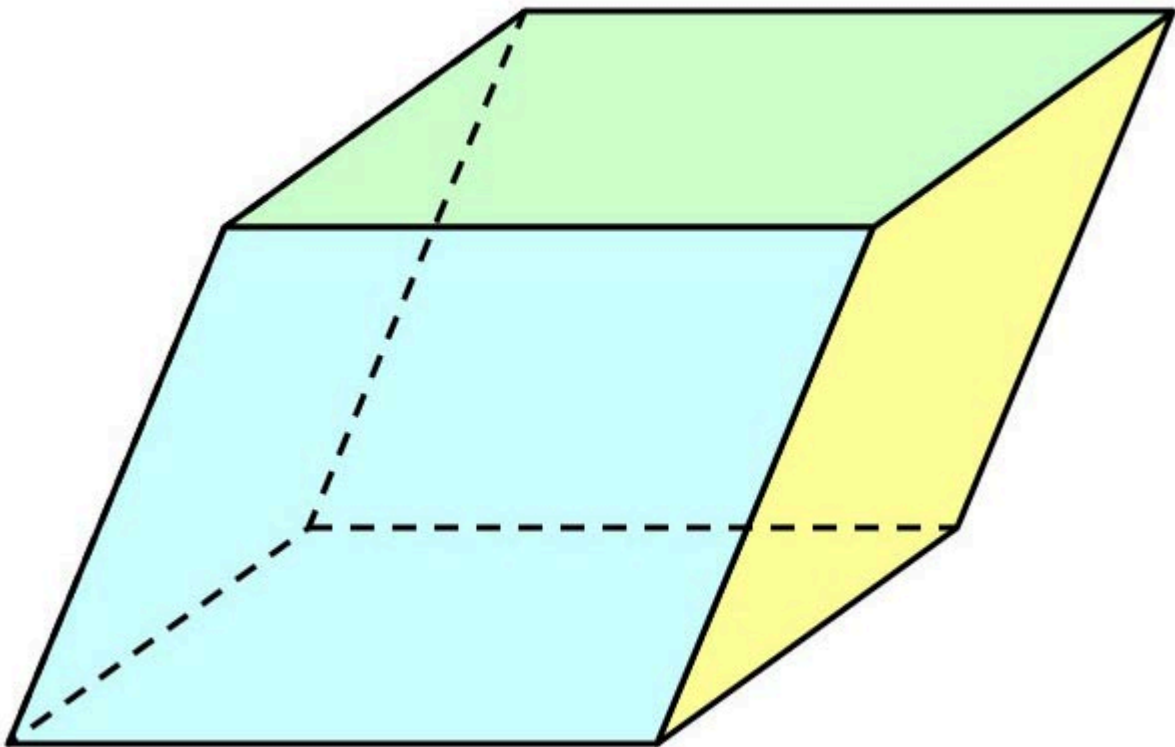
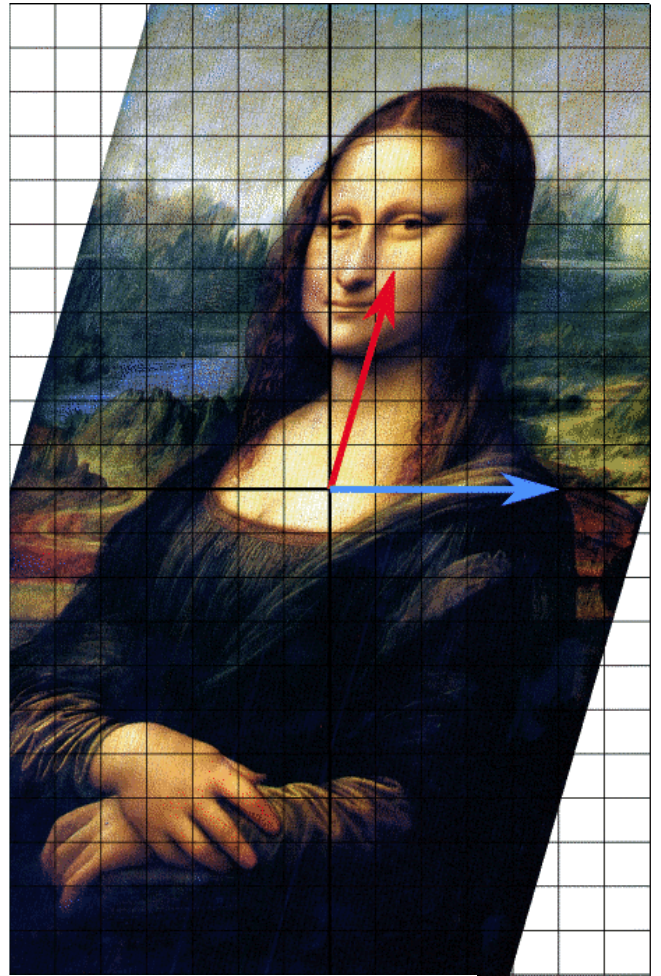
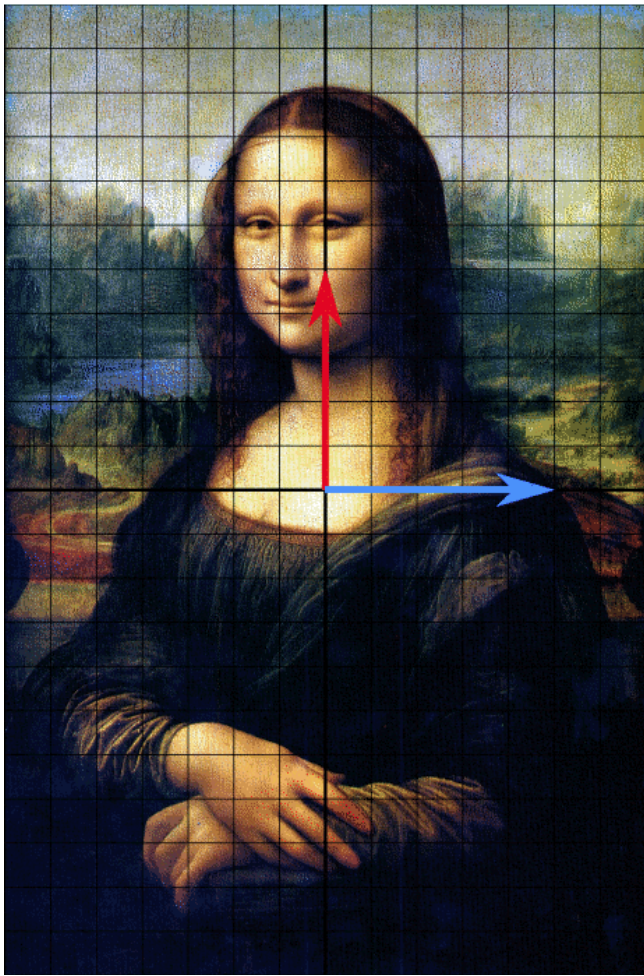
This is also why order matters: if you compose the matrices in one direction it might not be the same as the other.

$$\begin{matrix} 1 & 2 \\ 4 & 3 \end{matrix} \xrightarrow{\mathbf{V} \uparrow} \begin{matrix} 4 & 3 \\ 1 & 2 \end{matrix} \xrightarrow{\Theta_{90} \curvearrowright} \begin{matrix} 1 & 4 \\ 2 & 3 \end{matrix}$$

versus

$$\begin{matrix} 1 & 2 \\ 4 & 3 \end{matrix} \xrightarrow{\Theta_{90} \curvearrowright} \begin{matrix} 4 & 1 \\ 3 & 2 \end{matrix} \xrightarrow{\mathbf{V} \uparrow} \begin{matrix} 3 & 2 \\ 4 & 1 \end{matrix}$$

The actions can be composed (one after the other)—that's what multiplying matrices does. Eventually the matrix representing the overall cumulative effect of whatever things you composed, should be applied to something. For this you can say "the plane", or pick a few points, or draw an $\mathbf{F}$, or use a real picture (computers are good at linear transformations after all).



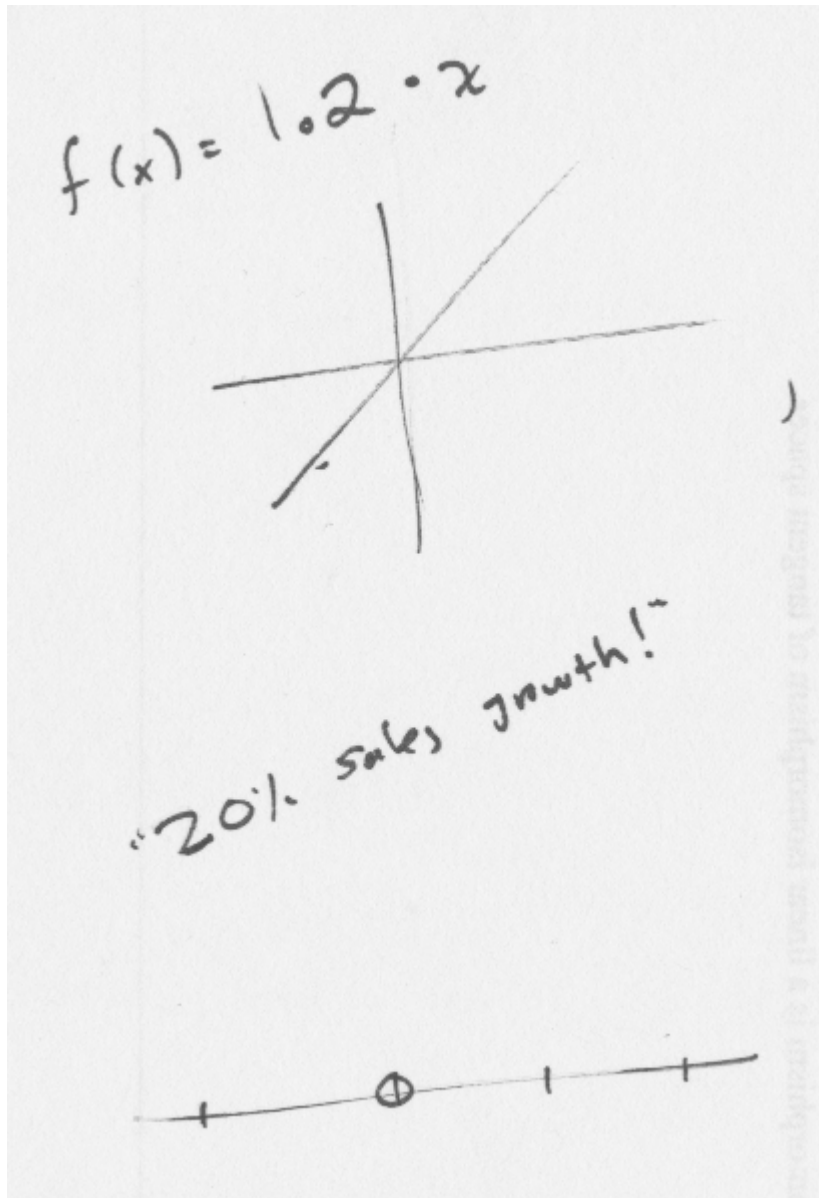
You could also watch the matrices work on Mona step by step too, to help your intuition.

Finally I think you *can* think of matrices as "multidimensional multiplication".

$$y = mx + b$$

is affine; the truly "linear" (keeping $0 \xrightarrow{f} 0$) would be less complicated: just

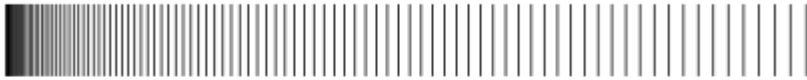
$$y = mx$$



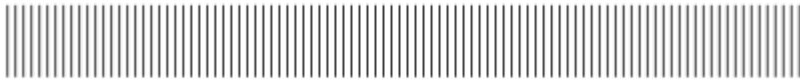
(eg.

) which is an "even"

3



2



1



stretching/dilation.

$$\vec{y} = [\mathbf{M}] \vec{x}$$

really is the multi-dimensional version of the same thing, it's just that when you have multiple numbers in each $\vec{x}$ each of the dimensions can impinge on each other

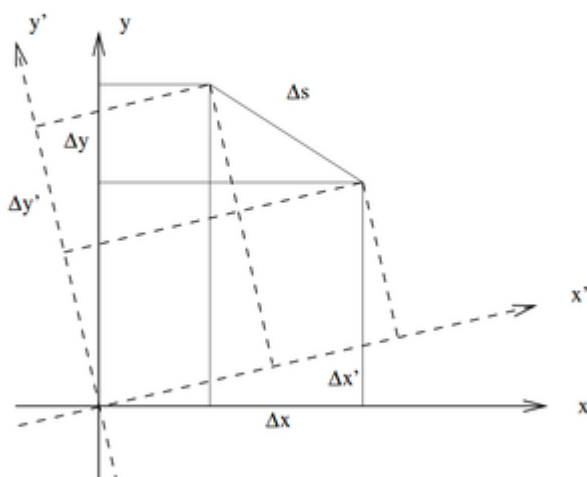
$$\left(\begin{array}{c|c|c} a \rightsquigarrow a & a \rightsquigarrow b & a \rightsquigarrow c \\ \hline b \rightsquigarrow a & b \rightsquigarrow b & b \rightsquigarrow c \\ \hline c \rightsquigarrow a & c \rightsquigarrow b & c \rightsquigarrow c \end{array} \right)$$

for example in the case of a rotation—in physics it doesn't matter which orthonormal coördinate system you choose, so we want to "quotient away" that invariant our physical theories.

In a different Cartesian coordinate system, defined by x' and y' axes which are rotated with respect to the originals, the formula for the distance is unaltered:

$$s^2 = (\Delta x')^2 + (\Delta y')^2. \quad (1.2)$$

We therefore say that the distance is invariant under such changes of coordinates.



This is why it is useful to think of the plane as 2-dimensional: although we use two distinct numbers to label each point, the numbers are not the essence of the geometry, since we can rotate axes into each other while leaving distances and so forth unchanged. In Newtonian

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edited Feb 10, 2019 at 22:10

answered Oct 3, 2014 at 5:39



Glorfindel

3,975

10

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38



isomorphismes

4,682

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53

4 what happened to the captions for parts (e) and (f) of the kazdan graphic? – MichaelChirico Oct 2, 2016 at 0:54



First, Understand Vector multiplication by a scalar.

18

Then, think on a Matrix, multiplied by a vector. The Matrix is a "vector of vectors".



Finally, Matrix X Matrix extends the former concept.



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answered Apr 8, 2011 at 22:56

hhh

291

1

4

This is a great answer. Now explained enough though. It is a similar way to how Gilbert Strang 'builds' up matrix multiplication in his video lectures. – Vass May 18, 2011 at 15:49

1 This is the best answer so far as you do not need to know about transformations. – matqkks Aug 6, 2013 at 20:14

2 The problem with this approach is to motivate the multiplication of a matrix by a vector. – Maxis Jaisi May 19, 2017 at 6:30



18



Apart from the interpretation as the composition of linear functions (which is, in my opinion, the most natural one), another viewpoint is on some occasions useful.

You can view them as something akin to a generalization of [elementary row/column operations](#). If you compute AB , then the coefficients in j -th row of A tell you, which linear combination of rows of B you should compute and put into the j -th row of the new matrix.

Similarly, you can view AB as making linear combinations of columns of A , with the coefficients prescribed by the matrix B .

With this viewpoint in mind you can easily see that, if you denote by $\vec{a}_1, \dots, \vec{a}_k$ the rows of the matrix A , then the equality

$$\begin{pmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_k \end{pmatrix} B = \begin{pmatrix} \vec{a}_1 B \\ \vdots \\ \vec{a}_k B \end{pmatrix}$$

holds. (Of course, you can obtain this equality directly from definition or by many other methods. My intention was to illustrate a situation, when familiarity this viewpoint could be useful.)

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edited Jan 5, 2017 at 9:14

answered Apr 8, 2011 at 13:28



Martin Sleziak

55.2k

20

203

381

- 1 I came here to talk about composition of linear functions, or elementary row/column operations. But you already said it all. Great job, and +1 also you would deserve more. – Wok Apr 8, 2011 at 21:01



15



Suppose

$$p = 2x + 3y$$

$$q = 3x - 7y$$

$$r = -8x + 9y$$

Represent this way from transforming $\begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} p \\ q \\ r \end{bmatrix}$ by the matrix

$$\begin{bmatrix} 2 & 3 \\ 3 & -7 \\ -8 & 9 \end{bmatrix}.$$

Now let's transform $\begin{bmatrix} p \\ q \\ r \end{bmatrix}$ to $\begin{bmatrix} a \\ b \end{bmatrix}$:

$$a = 22p - 38q + 17r$$

$$b = 13p + 10q + 9r$$

represent that by the matrix

$$\begin{bmatrix} 22 & -38 & 17 \\ 13 & 10 & 9 \end{bmatrix}.$$

So how do we transform $\begin{bmatrix} x \\ y \end{bmatrix}$ directly to $\begin{bmatrix} a \\ b \end{bmatrix}$?

Do a bit of algebra and you get

$$a = \bullet x + \bullet y$$

$$b = \bullet x + \bullet y$$

and you should be able to figure out what numbers the four $\bullet$ s are. That matrix of four $\bullet$ s is what you get when you multiply those earlier matrices. That's why matrix multiplication is defined the way it is.

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edited Jul 14, 2015 at 20:13

answered Jul 7, 2013 at 21:06



Michael Hardy

1

- 2 Perhaps a link to [the other answer](#) with some explanation of how it applies to this problem would be better than duplicating an answer. [This meta question](#), and those cited therein, is a good discussion of the concerns. – robjohn ♦ Jul 8, 2013 at 14:22 ✎
 - 1 I like this answer. As linear algebra is really about linear systems so this answer fits in with that definition of linear algebra. – matqkks Aug 6, 2013 at 20:16
- I'm stuck on how $a=22p-38q+17r$ and $b=13p+10q+9r$ are obtained. If you wish to explain this I'd really appreciate. – Paolo Jun 27, 2014 at 15:28
- @Guandalino : It was not "obtained"; rather it is part of that from which one obtains something else, where the four $\bullet$ s appear. – Michael Hardy Jun 27, 2014 at 16:38
- 1 That's what I thought you meant. If someone is telling you how to calculate 42×19 , getting 798, then it makes sense to wonder how 798 was calculated, but 42 and 29 were not somehow calculated; rather they were chosen as an example. The same is true of the numbers you mention. – Michael Hardy Jun 27, 2014 at 19:31



9



To get away from how these two kinds of multiplications are implemented (repeated addition for numbers vs. row/column dot product for matrices) and how they behave symbolically/algebraically (associativity, distribute over 'addition', have annihilators, etc), I'd like to answer in the manner 'what are they good for?'

Multiplication of numbers gives area of a rectangle with sides given by the multiplicands (or number of total points if thinking discretely).

Multiplication of matrices (since it involves quite a few more actual numbers than just simple multiplication) is understandably quite a bit more complex. Though the composition of linear functions (as mentioned elsewhere) is the essence of it, that's not the most intuitive description of it (to someone without the abstract algebraic experience). A more *visual* intuition is that one matrix, multiplying with another, results in the transformation of a set of points (the columns of the right-hand matrix) into new set of points (the columns of the resulting matrix). That is, take a set of n points in n dimensions, put them as columns in an $n \times n$ matrix; if you multiply this from the left by another $n \times n$ matrix, you'll transform that 'cloud' of points to a new cloud.

This transformation isn't some random thing; it might rotate the cloud, it might expand the cloud, it won't 'translate' the cloud, it might collapse the cloud into a line (a lower number of dimensions might be needed). But it's transforming the whole cloud all at once smoothly (near points get translated to near points).

So that is one way of getting the 'meaning' of matrix multiplication.

I have a hard time getting a good metaphor (any metaphor) between matrix multiplication and simple numerical multiplication so I won't force it - hopefully someone else might be able to come up with a better visualization that shows how they're more alike beyond the similarity of some algebraic properties.

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edited Sep 17, 2015 at 13:13

answered Apr 8, 2011 at 20:53



Mitch

8,737

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71



5

You shouldn't try to think in terms of scalars and try to fit matrices into this way of thinking. It's exactly like with real and complex numbers. It's difficult to have an intuition about complex operations if you try to think in terms of real operations.



scalars are a special case of matrices, as real numbers are a special case of complex numbers.



So you need to look at it from the other, more abstract side. If you think about real operations in terms of complex operations, they make complete sense (they are a simple case of the complex operations).

And the same is true for Matrices and scalars. Think in terms of matrix operations and you will see that the scalar operations are a simple (special) case of the corresponding matrix operations.

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edited Feb 21, 2016 at 23:24

answered Apr 8, 2011 at 17:38



John B

17.6k

11

27

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user9310



4



Let's explain where matrices and matrix multiplication come from. But first a bit of notation: e_i denotes the column vector in $\mathbb{R}^n$ which has a 1 in the i th position and zeros elsewhere:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation. The matrix

$$A = [T(e_1) \quad \cdots \quad T(e_n)]$$

is called the *matrix description* of T . If we are given A , then we can easily compute $T(x)$ for **any** vector

$x \in \mathbb{R}^n$, as follows: If

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

then

$$T(x) = T(x_1 e_1 + \cdots + x_n e_n) = x_1 \underbrace{T(e_1)}_{\text{known}} + \cdots + x_n \underbrace{T(e_n)}_{\text{known}}. \quad (1)$$

In other words, $T(x)$ is a **linear combination of the columns of A** . To save writing, we will denote this particular linear combination as Ax :

$$Ax = x_1 \cdot (\text{first column of } A) + x_2 \cdot (\text{second column of } A) + \cdots + x_n \cdot (\text{final column of } A).$$

With this notation, equation (1) can be written as

$$T(x) = Ax.$$

Now suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations, and that T and S are represented by the matrices A and B respectively. Let a_i be the i th column of A .

Question: What is the matrix description of $S \circ T$?

Answer: If C is the matrix description of $S \circ T$, then the i th column of C is

$$\begin{aligned} c_i &= (S \circ T)(e_i) \\ &= S(T(e_i)) \\ &= S(a_i) \\ &= Ba_i. \end{aligned}$$

For this reason, we define the "product" of B and A to be the matrix whose i th column is Ba_i . With this notation, the matrix description of $S \circ T$ is

$$C = BA.$$

If you write everything out in terms of components, you will observe that this definition agrees with the usual definition of matrix multiplication.

That is how matrices and matrix multiplication can be discovered.

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answered Dec 24, 2018 at 7:21



littleO

53.3k

8

103

179



Let's say you're multiplying matrix A by matrix B and producing C : $A \times B = C$

3

At a high level, A represents a set of linear functions or transformations to apply, B is a set of values for the input variables to run through the linear functions/transformations, and C is the result of applying the linear functions/transformations to the input values.

For more details, see below:

(1) A consists of rows, each of which corresponds to a linear function/transformation

For instance, let's say the first row in A is $[a_1 \ a_2 \ a_3]$ where a_1 , a_2 and a_3 are constants. Such row corresponds to the linear function/transformation:

$$f(x, y, z) = a_1x + a_2y + a_3z$$

(2) B consists of columns, each of which corresponds to values for the input variables.

For instance, let's say the first column in B consists of the values $[b_1 \ b_2 \ b_3]$ (transpose). Such column correspond to setting the input variable values as follows:

$$x = b_1$$

$$y = b_2$$

$$z = b_3$$

(3) C consists of entries whereby an entry at row i and column j corresponds to applying the linear function/transformation represented in A 's i th row to the input variable values provided by B 's j th column.

Thus, the entry in the first row and first column of C in the example discussed thus far, would equate to:

$$\begin{aligned} f(x = b_1, y = b_2, z = b_3) &= a_1x + a_2y + a_3z \\ &= a_1b_1 + a_2b_2 + a_3b_3 \end{aligned}$$

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edited Oct 2, 2016 at 1:01

answered Oct 21, 2015 at 19:57



MichaelChirico

4,221 1 17 29



blu

131 1



2



One way to try to "understand" it would be to think of two factors in the product of two numbers as two different entities: one is *multiplying* and the other is *being multiplied*. For example, in $5 \cdot 4$, 5 is the one that is multiplying 4, and 4 is being multiplied by 5. You can think in terms of repetitive addition that you are adding 4 to itself 5 times: You are doing *something* to 4 to get another number, that *something* is characterized by the number 5. We forbid the interpretation that you are adding 5 to itself 4 times here, because $5 \cdot$ is an *action* that is related to the number 5, it is not a number. Now, what happens if you multiply 4 by 5, and then multiply the result by 3? In other words,

$$3 \cdot (5 \cdot 4) = ?$$

or more generally,

$$3 \cdot (5 \cdot x) = ?$$

when we want to vary the number being multiplied. Think of for example, we need to multiply a lot of different numbers x by 5 first and then by 3, and we want to somehow simplify this process, or to find a way to compute this operation fast. Without going into the details, we all know the above is equal to

$$3 \cdot (5 \cdot x) = (3 \times 5) \cdot x,$$

where 3×5 is just the ordinary product of two numbers, but I used different notation to emphasize that this time we are multiplying "multipliers" together, meaning that the result of this operation is an entity that multiplies other numbers, in contrast to the result of $5 \cdot 4$, which is an entity that gets multiplied. Note that in some sense, in 3×5 the numbers 5 and 3 participates on an equal ground, while in $5 \cdot 4$ the numbers have differing roles. For the operation $\times$ no interpretation is available in terms of repetitive addition, because we have two actions, not an action and a number. In linear algebra, the entities that gets multiplied are *vectors*, the "multiplier" objects are *matrices*, the operation $\cdot$ generalizes to the *matrix-vector product*, and the operation $\times$ extends to the *product between matrices*.

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edited Apr 8, 2011 at 18:13

answered Apr 8, 2011 at 17:57



timur

16.9k

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Instead of trying to explain it, I will link you to an interactive graphic. As you play around with it, watch how the matrix in the top left corner changes as you move X' and Y'. The intuition for how matrix multiplication works comes after the intuition for where it's used.

<https://www.geogebra.org/m/VjhNaB8V>

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answered May 10, 2017 at 12:25



Nathan Petrangelo

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This is the detailed and easiest explanation :

<https://nolaymanleftbehind.wordpress.com/2011/07/10/linear-algebra-what-matrices-actually-are/>

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answered Feb 7, 2019 at 3:01



Nikhil Chilwant

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